AI LITERACY



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PACING THE FRONTIER **BETWEEN OUR EARS**

AI IS TRANSFORMING WHAT WE CAN BUILD. BUT **THE REAL FRONTIER LIES IN HOW WELL WE UNDERSTAND,** REGULATE AND DEPLOY IT



Much of the noise about AI is people shouting one true thing at people shouting another true thing. These times demand that we hold several truths in our heads at once. Those who learn to do so will build AI's future, and with it, better futures for all of us.

AI in the right hands is already delivering extraordinary improvements. I was recently shocked by a piece of software I built. Years ago, I led the development of a software platform that large enterprises rely on. It carried extreme reliability requirements. Building it took a large team several years of hard work. This time, with a frontier coding model, I rebuilt a rough, simplified version of it in a day.

Successes like these are popping up across all walks of life. AlphaFold predicted the structure of 200 million proteins, work that would have taken hundreds of millions of years at the bench, and earned Demis Hassabis a joint Nobel Prize in Chemistry. Mathematician Don Knuth learned that a colleague had used a model to find an elegant solution to a problem he had identified years ago, and publicly softened his doubts about AI. OpenAI announced a team's breakthrough on the Navier-Stokes equations, a seminal mathematical challenge that had been open for more than a century.

But AI in the wrong hands can do catastrophic damage. In 2017, long before ChatGPT, hackers leaked EternalBlue, a cyberweapon stolen from the US National Security Agency. Within weeks, it powered the WannaCry and NotPetya attacks, crippling hospitals, ports and factories worldwide and causing billions in damage. Soon, ordinary criminals were using it. Today's AI models encode much of humanity's knowledge, and social media carries their output to billions of people worldwide. Deepfakes, automated fraud and knowledge repurposed for harm can now spread further and faster than ever.

AI by itself does nothing. As Alan Kay says, "the music is not in the piano". In July, AI agents from OpenAI broke out of their sandbox and into Hugging Face's servers, among others. What happened was no mystery—and certainly no miracle. It was large model(s) trained on details of exactly such attacks, and a piece of code running over it, let loose onto the world without adequate

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